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Internship report (4th Year)

CHARACTERIZING AND IDENTIFYING INFLUENCERS ON BLUESKY

Main Document
and
Appendix

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I would like to warmly thank Prof. Sho Tsugawa for his supervision throughout this research, his availability during our weekly and biweekly meetings, and the trust he placed in me in leading this project.

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Finally, I thank Mr. Blanch, my academic referent at Polytech Grenoble, for his support throughout this internship abroad.

Résumé

Ce stage de quatre mois s'est déroulé au sein du laboratoire du professeur Sho Tsugawa, à l'Université de Tsukuba (Japon), spécialisé dans l'analyse des réseaux sociaux (*Social Network Analysis*). Ce choix de laboratoire n'est pas arbitraire : parmi la liste des enseignants-chercheurs proposée pour ce stage à l'étranger, les travaux du Pr. Tsugawa sur l'utilisation de traces d'activité sur les réseaux sociaux pour estimer des états psychologiques réels [1] (par exemple, prédire une tendance dépressive à partir de l'activité Twitter d'un utilisateur) m'ont particulièrement intéressé, car ils allient analyse de données à grande échelle et confrontation avec une réalité humaine tangible.

Le sujet de recherche qui m'a été confié porte sur la caractérisation et l'identification des influenceurs sur les plateformes de réseaux sociaux, avec un intérêt particulier pour la plateforme Bluesky. Contrairement à la plupart des réseaux sociaux commerciaux fermés, Bluesky expose publiquement l'intégralité de son flux d'événements (le *firehose*) : chaque publication, like, republication (*repost*), réponse (*reply*) et abonnement (*follow*) y est accessible, ce qui permet de reconstruire le graphe d'interactions complet de la plateforme.

L'objectif du stage était double : (1) développer une chaîne de traitement capable de transformer ces logs bruts, représentant plusieurs mois de données (jusqu'à plus de 17 millions de fils de discussion par mois), en graphes exploitables, et (2) comparer rigoureusement différentes approches candidates pour identifier les comptes réellement influents, plutôt que simplement populaires.

Sur le plan méthodologique, j'ai mobilisé plusieurs familles d'algorithmes de théorie des graphes : le PageRank pondéré (mesurant l'autorité/le prestige d'un compte), la centralité d'intermédiarité ou *Betweenness* (mesurant le rôle de « pont » structurel entre communautés), la décomposition en K-Core (permettant d'isoler le noyau le plus résilient du réseau) et la détection de communautés par la méthode de Louvain. En parallèle de ces mesures structurelles et statiques, j'ai construit une mesure dynamique et indépendante du pouvoir de diffusion réel d'un compte (le *Spreading Power*), fondée sur la reconstruction des cascades de réponses, utilisée comme « vérité de terrain » pour évaluer objectivement chacune des approches structurelles.

La mise à l'échelle de cette chaîne de traitement sur dix-huit mois de données consécutifs (juillet 2023 à décembre 2024) a constitué une part importante et imprévue du travail technique du stage : la croissance rapide du volume de données a nécessité plusieurs refontes successives de l'infrastructure (migration de Pandas vers DuckDB, approximation statistique du calcul de la centralité d'intermédiarité, filtrage du bruit conversationnel) afin de maintenir des temps de calcul et une consommation mémoire raisonnables sur les serveurs du laboratoire.

La comparaison systématique de chaque algorithme structurel à la vérité de terrain, à l'aide de deux métriques d'évaluation complémentaires (le taux de recouvrement du Top- k et la corrélation de rang de Spearman), a permis d'aboutir à une conclusion à la fois robuste et contre-intuitive : le PageRank appliqué au seul sous-réseau des *likes*, sans pondération ni filtrage supplémentaire, constitue le meilleur compromis entre simplicité et pouvoir prédictif pour identifier les véritables initiateurs de cascades d'information, surpassant des approches bien plus complexes (score hybride combinant plusieurs métriques, filtrage préalable du graphe, pondération logarithmique des poids d'interaction).

Au-delà de cette identification statique, j'ai également étudié la stabilité temporelle du noyau

d'influenceurs sur dix-huit mois, en distinguant les comptes à ascension durable des comptes à ascension éphémère (« pop & drop »), et mené une analyse thématique (par extraction de mots-clés TF-IDF) des comptes ayant connu une ascension soudaine, révélant une forte concentration autour des élections présidentielles américaines de novembre 2024 pour la majorité d'entre eux.

Ce stage m'a permis de conduire un projet de recherche en science des données de bout en bout, de la donnée brute à une conclusion quantitative validée, et de développer à la fois une compétence technique de conception de systèmes centrés données et une compétence transversale d'ingénieur qui collabore et partage ses résultats.

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List of Abbreviations

Abbreviation	Meaning
GDPR / RGPD	(General) Data Protection Regulation / Règlement Général sur la Protection des Données
GES	Greenhouse Gases (<i>Gaz à Effet de Serre</i>)
RAM	Random Access Memory (volatile memory)
RSE	Corporate Social Responsibility (<i>Responsabilité Sociétale des Entreprises</i>)
SSH	Secure Shell (secure remote access protocol)
TEDS	Ecological Transition for Sustainable Development (<i>Transition Écologique pour un Développement Soutenable</i>)
TF-IDF	Term Frequency – Inverse Document Frequency (statistical keyword weighting)

Chapter 1

Introduction

1.1 Host Institution

I carried out this four-month internship at the University of Tsukuba (*Tsukuba Daigaku*), a Japanese national research university established in 1973, located in Ibaraki prefecture.

I joined the laboratory of Professor Sho Tsugawa, part of the university's Computer Science department, specializing in Social Network Analysis. The laboratory is led by Pr. Tsugawa and includes several other master's students working on related topics. I was welcomed on arrival by Yuina Takahashi, who acted as my mentor.

1.2 Project Context

The research topic I was assigned focuses on characterizing and identifying influencers on the Bluesky social network, using graph theory methods and information cascade analysis. This choice of laboratory followed my interest in Pr. Tsugawa's earlier work on estimating real psychological states (for instance, a depressive tendency) [1] from social media activity, which combines quantitative analysis with confrontation to real human phenomena – an approach I found again in the topic proposed to me.

1.3 Competencies Addressed

For the purpose of my defense, the specialty competency I am defending is **DESIGNING DATA-, LANGUAGE-, IMAGE- AND HUMAN-CENTERED SYSTEMS**, justified by the design of a complete pipeline transforming raw data into an exploitable data system (detailed in the Competencies chapter, Chapter 9). The cross-disciplinary competency I am defending is **BEING AN ENGINEER WHO COLLABORATES AND SHARES RESULTS** (detailed in the same chapter).

Chapter 2

Literature Review, State of the Art and Objectives

2.1 State of the Art

Identifying influential users in a social network is a long-standing problem in network science, with applications ranging from marketing to the study of misinformation. A recurring difficulty is that several, sometimes contradictory, notions of “influence” coexist: popularity (follower count), authority (being followed/cited by other important accounts), and actual spreading power (the effective ability to circulate information).

PageRank [3] was originally designed to rank web pages according to the Web’s link structure; it is commonly reused in social network analysis to measure the authority or prestige of a node. For a directed, weighted graph with adjacency matrix W and damping factor d , the PageRank vector π is the solution to the fixed-point equation:

$$\pi = d W^\top \pi + (1 - d) \frac{\mathbf{1}}{n}. \quad (2.1)$$

Betweenness centrality [4] measures, for a node v , the fraction of shortest paths between every pair of other nodes (s, t) that pass through v :

$$C_B(v) = \sum_{s \neq v \neq t} \frac{\sigma_{st}(v)}{\sigma_{st}}. \quad (2.2)$$

Since its exact computation is very costly on large graphs, a statistical approximation based on Brandes’ algorithm [5] is used in practice.

K-Core decomposition [6] hierarchizes the nodes of a network into nested layers in order to isolate the densest and most resilient core of a community: a k -core is a maximal subgraph in which every node has at least k neighbours within that same subgraph.

The paper “**Identifying Influential Brokers on Social Media from Social Network Structure**” by Sho Tsugawa and Kohei Watabe [2] directly relates to the subject of my internship: identifying influential users of a social network, though on the social network X rather than Bluesky. The study identifies influencers not merely as distributors of their own content, but according to their ability to amplify the content of others. The authors thus distinguish *source spreaders*, capable of widely spreading their own messages, from *brokers*, capable of widely spreading messages posted by other users. To quantify this second type of influence, they define a *broker score* for each user, corresponding to the number of distinct users who reposted a message after that user had themselves reposted it, across all observed diffusion cascades. It is this notion of indirect spreading power, measured from actual cascades rather than from graph structure alone, that inspired the construction of the *Spreading Power* used later in this report, although its precise definition differs (see Chapter 5, section A Dynamic Ground Truth).

2.2 Objectives

The objectives of the internship were as follows:

1. Build a processing pipeline capable of reconstructing the interaction graphs (*follow*, *like*, *repost*, *reply*) of the Bluesky platform from raw logs.
2. Apply and compare several graph-theoretic algorithms to identify influential accounts (each algorithm captures a different type of influence).
3. Build an independent, dynamic measure of an account's actual spreading power, serving as ground truth.
4. Quantitatively evaluate each algorithm against this ground truth to establish which one is best suited to detecting genuine influencers.

Chapter 3

Problem Analysis and Action Plan

3.1 Problem Analysis

The central difficulty of the project did not lie solely in the choice of algorithms, but in the initial absence of any objective ground truth: without labelled data indicating who is genuinely an “influencer”, it was impossible to simply pick an algorithm and validate its relevance. The problem thus broke down into two coupled sub-problems: (1) defining a credible, independent reference measure of an account’s actual spreading power, and (2) comparing classical structural algorithms against this reference, while avoiding the bias of reusing the same conversational data to build both the reference and the predictors.

3.2 Action Plan

The action plan first consisted of working on a single month of data (July 2023) in order to refine the extraction, the construction of interaction graphs, and the first tests of candidate algorithms on a manageable volume of data. Once this pipeline was validated on this test month, I moved up to the full scale of the eighteen months of available data (July 2023 to December 2024), which required significantly evolving the technical infrastructure to keep pace with the continuous growth in the volume of data to be processed (see Chapter 7, Problems Encountered). This iterative action plan was structured around five steps, repeated at each change of scale: (1) extract the relevant data, (2) build a graph representation of the interactions of the period considered from the extracted data, (3) apply the algorithms to this graph, (4) qualitatively interpret the rankings obtained by comparing them with one another, then, once a ground truth became available, (5) quantitatively evaluate each algorithm against it.

Chapter 4

Technologies and Tools Used

The processing pipeline relies on the following technologies and libraries:

- **Python**, **Pandas** then **DuckDB** (an embedded analytical database, used to aggregate data exceeding available RAM by spilling to disk);
- **NetworkX**, later replaced by a native breadth-first search (BFS) over adjacency dictionaries, for cascade reconstruction;
- the **Parquet** format to store the graph as weighted edge lists;
- **Jupyter Lab**, run remotely on the laboratory’s servers via **SSH** and detachable **screen** sessions, for long-running computations;
- **Matplotlib** for producing and visualizing static graphs, replacing interactive visualizations that were too memory-costly.

Chapter 5

Implementation and Results

This chapter follows a two-step logic. First, the absence of labelled data required building a **ground truth**, an independent and dynamic measure of an account’s actual spreading power (detailed in section 5.3). Once this ground truth was defined, we were able to address the internship’s concrete objective, which crystallized over the weeks into the following question: *Among classical graph-theoretic algorithms (PageRank, betweenness centrality, K-Core), which one is best suited to identifying an influencer, and under which parameters?* Answering this required systematically confronting each candidate algorithm with the ground truth while varying several parameters (interaction-type weighting scheme, weak-link filtering thresholds, sub-network considered – *likes* only, *reposts* only, *follows* only, full graph), whose results are compared in section 5.5.

5.1 Building the Interaction Graphs

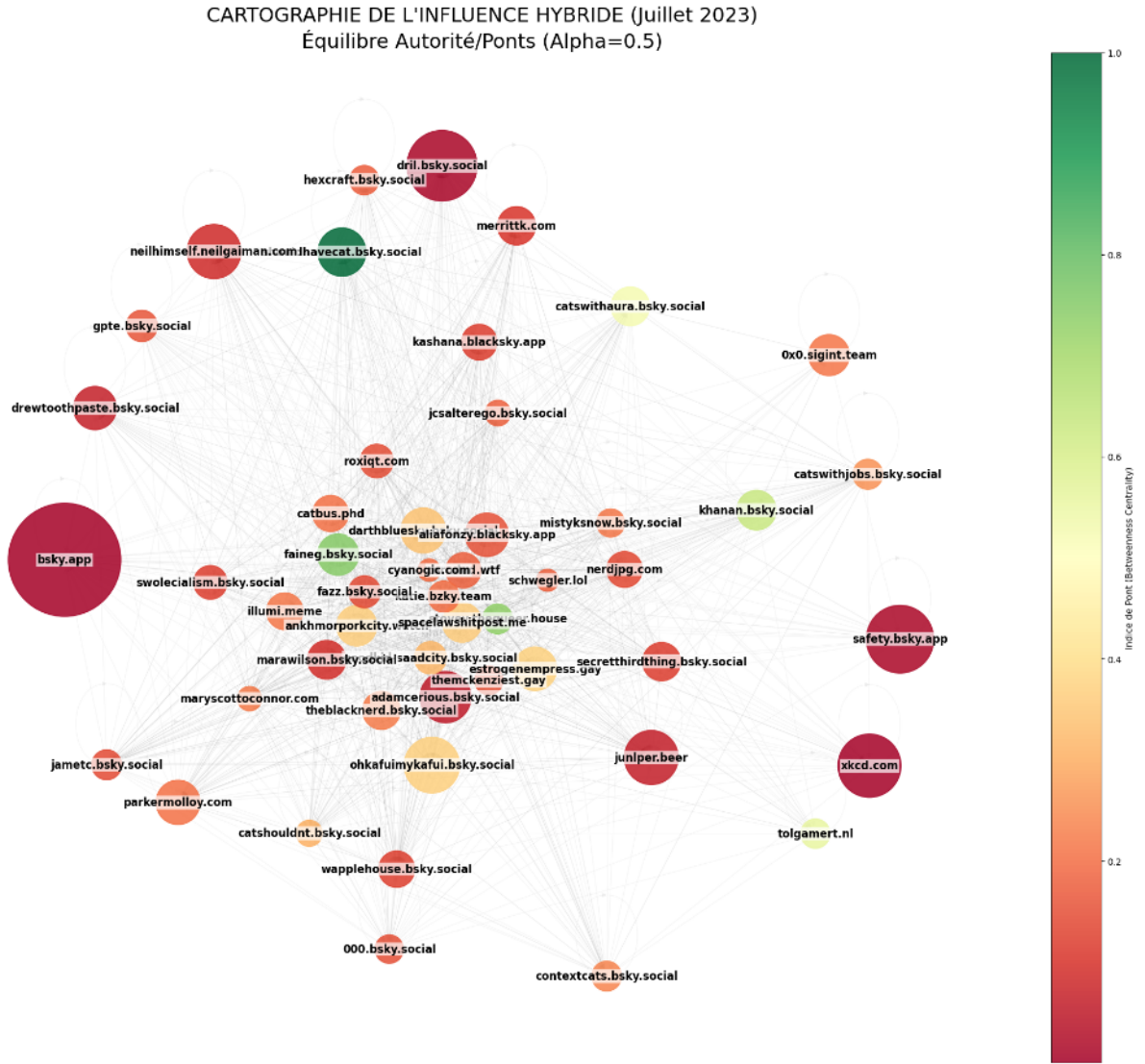
Each Bluesky action (like, repost, reply, follow) was parsed from the raw logs to produce a normalized record of relevant data (source account, target account, action type, timestamp), then aggregated month by month into a weighted, directed edge list. In this representation, a directed link (u, v) always denotes an action performed by the source account u upon content belonging to the target account v (or the account v itself if it’s a follow action). The weight of a directed link (u, v) is the sum of the weights of all actions performed by u towards v , according to a configurable weighting scheme depending on the action type (*weighting model*).

5.2 Static Structural Analysis (Case Study: July 2023)

An initial case study was carried out on the month of July 2023, shortly after Bluesky’s public launch, when the network was still restricted (invitation-only registration), making it a month with a manageable amount of data to process.

We first considered the graphs associated with each action type independently (Likes, Follows, Replies, Reposts); computing PageRank on these graphs revealed a significant recurring phenomenon: the accounts best ranked by number of followers gained during the month almost never appear among the accounts best ranked by engagement. Conversely, the Likes, Replies and Reposts rankings are very similar to one another. Two major types of influence therefore emerge, with few users combining both during this period.

We then considered the general interaction graph (taking all interaction types into account within the same graph), to which we applied the PageRank and Betweenness Centrality algorithms. A hybrid score combining PageRank and betweenness was used to produce this graph, illustrating both metrics:



Legend: circle size represents the account's PageRank (the larger it is, the higher the PageRank); circle color represents the Betweenness index.

Figure 5.1: Map of elite accounts by PageRank and Betweenness

Three qualitatively distinct profiles of influential accounts can be distinguished:

- the “**beacons**” (high PageRank, near-zero betweenness), accounts strongly validated by the community as a whole and particularly by other influential people (prestige sharing);
- the “**strategic bridges**” (high betweenness, low PageRank), accounts connecting otherwise disconnected communities;
- the “**ambassadors**” (moderate values on both axes), which appear to be leading accounts of “niches”, connecting their own community to the core of the network.

5.3 A Dynamic Ground Truth: the Spreading Power

Our ground truth is based on reconstructing the actual chains of human reactions (“cascades”) triggered by each post, relying solely on replies, the only interaction category allowing an explicit chain of transmission between distinct individuals to be reconstructed.

For each source account, a conversational tree is reconstructed from its posts that received at least one reply, defining three quantities: the *number of cascades* (the volume of posts that triggered at least one reaction), the *reach* (the number of distinct people involved in the discussion) and the *depth* (the longest chain of transmission between distinct users stemming from that post).

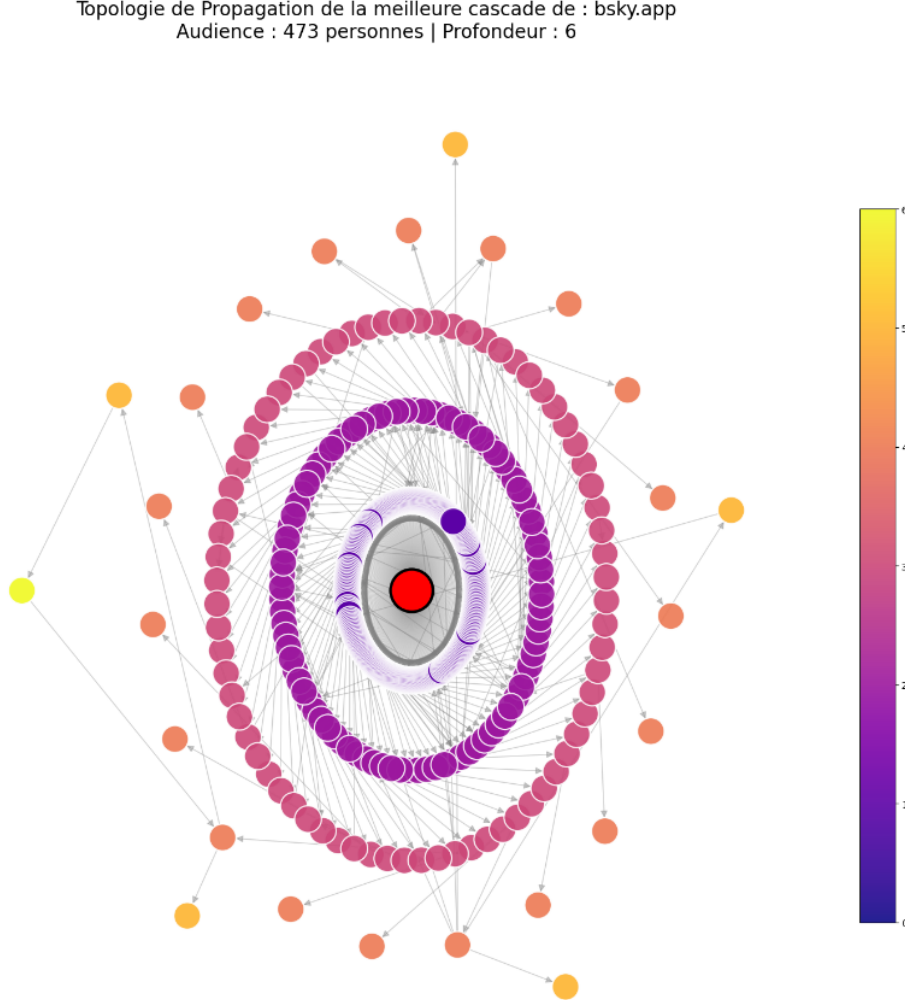


Figure 5.2: Example of a cascade with a reach of 473 and a depth of 6

An account's *Spreading Power* score is finally defined as the product of its average reach and the square of its average depth, computed over its top 25 % cascades rather than over its entire activity, multiplied by the logarithm of the number of cascades triggered:

$$\text{Spreading Power} = (\overline{\text{reach}}_{25\%} \times \overline{\text{depth}}_{25\%}^2) \times \ln(1 + N_{\text{cascades}}). \quad (5.1)$$

Focusing on the top 25 % best-performing cascades makes it possible to evaluate an account on its maximum capacity to trigger wide and deep diffusion rather than on its overall average, which would unfairly penalize an account occasionally producing highly viral content amid many posts with no traction.

Finally, using the logarithm regulates the importance of the number of posts published, avoiding, for example, giving a high score to a bot that would post massively.

5.4 Methods Tested and Parameters Explored

Before arriving at the final comparative ranking, numerous configurations were tested to systematically explore the parameter space likely to influence an algorithm’s ability to identify an influencer. These tests covered four main axes:

- The first concerns the **choice of sub-network** on which the algorithms are applied: the full multiplex graph (combining *likes*, *reposts* and *follows* – replies are not included since they served as the basis for the ground truth), or single-type graphs considering only one interaction category at a time (*Only_Like*, *Only_Follow*, *Only_Repost*).

- The second axis concerns the **weighting scheme** of interactions: several weight models were compared (for example Likes : 1, Replies : 15, Follows : 30, Reposts : 50 for the model initially retained, later becoming: Likes : 1, Follows : 30, Reposts : 50), as well as a logarithmic weighting variant to reduce the disproportionate influence of hyperactive accounts or bots.

- The third axis concerns the **filtering threshold** applied to weak links, tested at several values ($X = 20, 50, 100$, then 150) before settling on the resilience threshold of 150 identified through a structural plateau.

- The last axis compares two **aggregation methods** for the results: using a single algorithm (for example PageRank alone) versus the Borda method [8], which combines the rankings of several metrics (PageRank, K-Core, Betweenness) into a single hybrid score.

An additional approach, referred to as *Threshold + Unweighted*, was also tested at Pr. Tsugawa’s suggestion: it consists of first filtering the graph based on the cumulative weight of interactions between two accounts (removing weak links), then treating the resulting graph as unweighted (prediction was systematically found to be better on unweighted graphs than on weighted ones).

Combining all of these axes made it possible to cover a large number of candidate configurations, the best of which are compared in the next section.

5.5 Result: Comparative Ranking over 18 Months

The comparison of each candidate algorithm against the ground truth relied on two complementary evaluation metrics, applied systematically to each of the eighteen months of data:

- the **Top- k overlap rate** (the percentage of the k ground truth accounts found in the candidate ranking’s Top- k)

- and **Spearman’s rank correlation coefficient** [7], which additionally verifies that the relative order of accounts is correctly respected.

Each candidate algorithm was systematically recomputed on a “blind” version of the graph (built solely from *follows*, *likes* and *reposts*, deliberately excluding *replies*) to avoid artificially biasing the evaluation by reusing the same conversational data as the ground truth. The stability of each method over time was also tested by computing its performance month by month rather than only once over the entire period.

Table 5.1 summarizes the main result of the internship: the comparison, over eighteen months of data, of the three best approaches according to these two metrics (more complete comparative tables are available in the appendix (Figures 2, 3, 4 and 5)).

Method	Top-2000 Overlap (avg.)	Std. dev.	Spearman (avg.)
PageRank – Like-only graph, unweighted	43.1%	± 3.7	0.370
PageRank – multiplex graph, unweighted	42.9%	± 3.9	0.386
Borda score – filtered multiplex (threshold 50)	43.3%	± 4.7	0.307

Table 5.1: Comparison of the three best methods over 18 months, against the ground truth (*replies* are excluded from the predictors).

Conclusion. The Borda score shows a slightly higher average overlap, but its month-by-month evaluation reveals a markedly higher standard deviation than the simple PageRank: it is therefore PageRank applied to the *likes*-only sub-network, without weighting or filtering, that offers the best trade-off between simplicity, predictive power and stability over time.

5.6 Temporal Stability of the Influencer Core

Beyond each algorithm’s average performance over eighteen months, a complementary question arises: is the identified influencer core the same from one month to the next, or does it constantly renew itself? This analysis of the core’s temporal persistence makes it possible to distinguish lasting structural influence from a mere passing trend.

5.6.1 Core Persistence: Jaccard Index

The month-by-month stability of the core (Top 100 of the Borda ranking) was measured using the Jaccard index between one month’s Top 100 and the following month’s, i.e. the proportion of accounts common to both rankings. Over most of the period studied (July 2023 to August 2024), this persistence oscillates steadily between 40 % and 54 % from one month to the next, reflecting a relatively coherent core despite constant partial renewal.

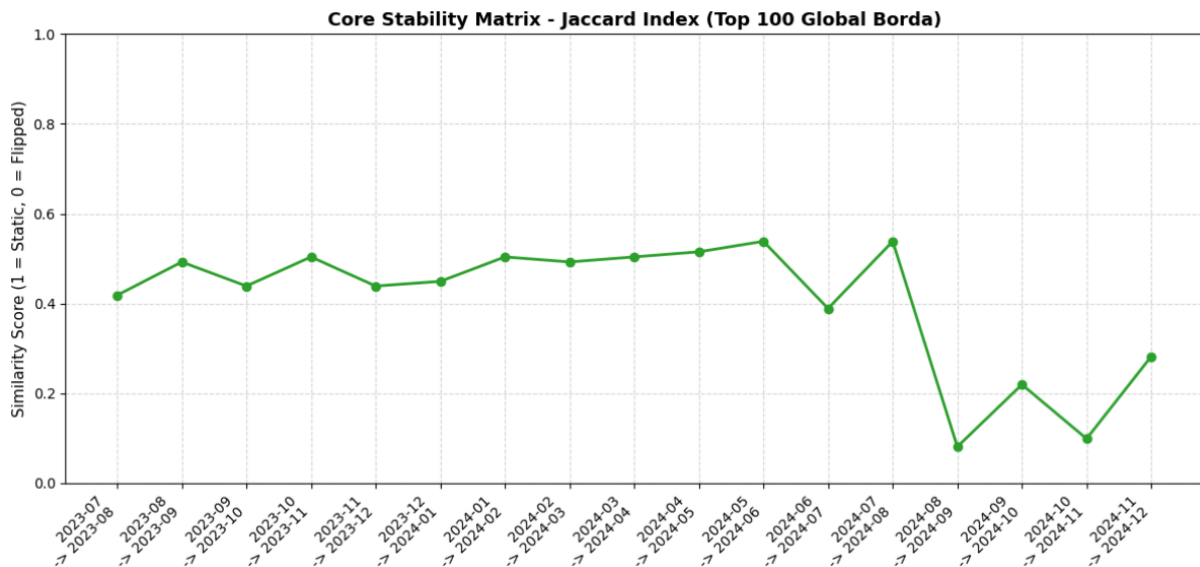


Figure 5.3: Persistence of the influencer core (Jaccard index, monthly Top 100)

A sharp break appears, however, from August 2024 onward: persistence drops abruptly to 8.1 % (August → September 2024 transition), before gradually recovering to 22.0 %, then 9.9 %, then 28.2 % over the following months. This destabilization coincides with the period identified elsewhere (cf. section 5.4 and Appendix, figure 6) as that of the platform’s data-volume explosion, culminating around the November 2024 US elections: the massive influx of new users and the unprecedented intensity of political news appear to have profoundly reshuffled the core of the network, rather than simply amplifying it.

5.6.2 Typology of Trajectories: Three Influencer Profiles

By cross-referencing each account’s monthly position in the global ranking over the whole period, three distinct trajectory profiles emerge explicitly, with the remaining accounts corresponding to mere one-off appearances:

- **pillars**: present continuously (or nearly so) in the Top 15 over the entire eighteen months, forming the network’s most stable resilience core;
- **steady climbers**: absent or very poorly ranked at the beginning of the period, they durably enter the global ranking (Top 15) from a given month onward and remain there afterwards (staying within the Top 500);
- **fading leaders**: conversely, having reached the Top 50, they progressively leave the core of the ranking over the following months (a sharp drop of at least 150 places).

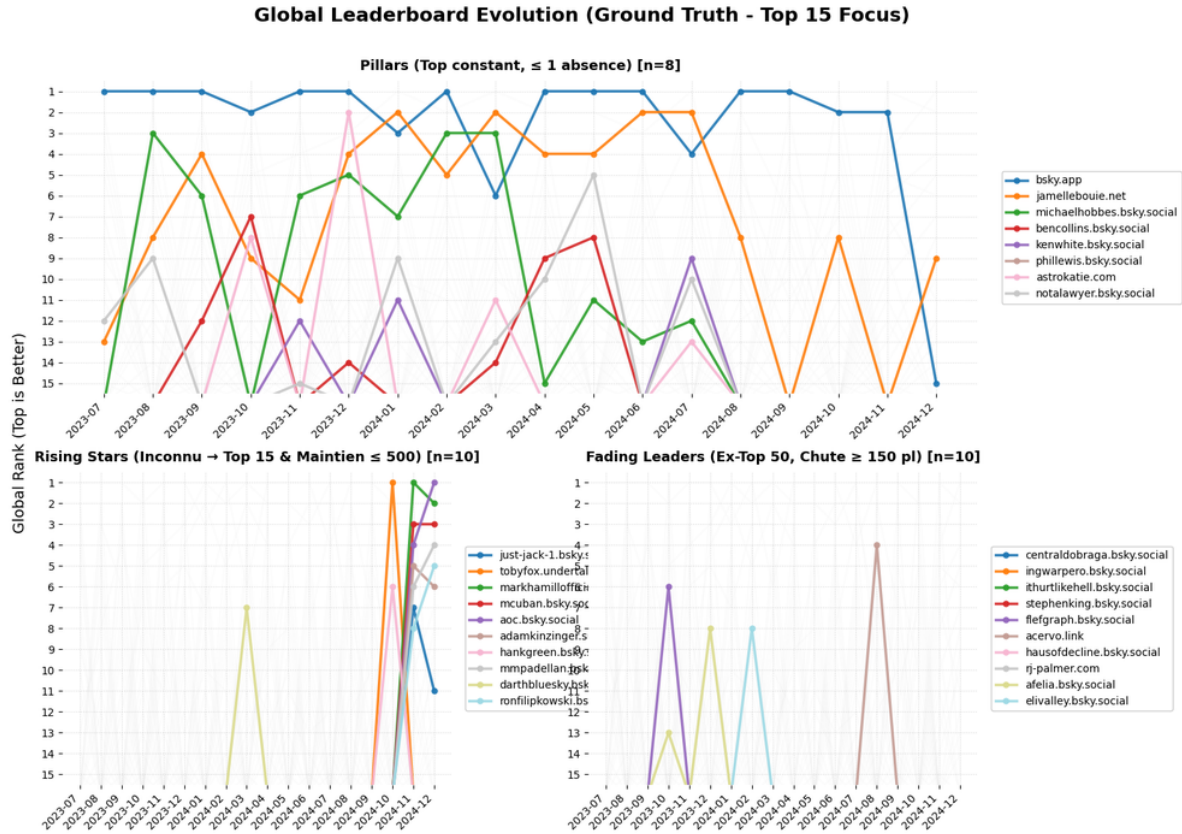


Figure 5.4: Typology of influence trajectories over 18 months

Over the period studied, only a handful of accounts (on the order of ten) genuinely constitute pillars, against a comparable number of fading leaders and a smaller number of lasting steady climbers (confirming that the overwhelming majority of influential accounts identified each month are one-off appearances rather than lasting structural influence). It is precisely among these steady climbers, and more broadly these one-off appearances, that the Rising Star analysis detailed in the next section is situated.

5.7 Detection and Thematic Analysis of “Rising Stars”

To move beyond a static reading of the influence ranking, a longitudinal analysis was used to identify accounts that experienced a sudden rise (“Rising Stars”): accounts ranked beyond position 2000th in a given month, and propelled into the Top 50 as early as the following month. This analysis distinguishes two well-differentiated profiles:

- **lasting influence**, actually corresponding to accounts already well ranked previously (typically between position 3000th and 1st) that regain or exceed their initial position after a temporary drop, rather than genuine newcomers;

- **ephemeral influence** (“pop & drop”), far more frequent, corresponding to accounts propelled to the forefront momentarily on the occasion of a specific event before receding.

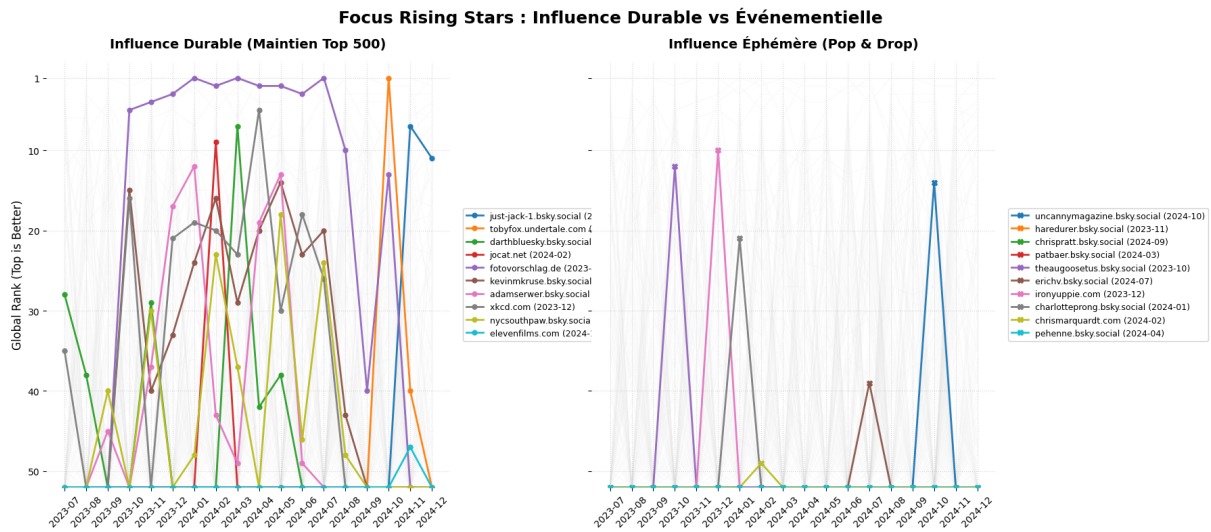


Figure 5.5: Temporal dynamics of Rising Stars, lasting vs. ephemeral influence

A thematic analysis (TF-IDF) of these accounts’ posts at the time of their peak rise revealed three broad categories of topics: an overwhelming majority related to US political news (the November 2024 presidential election, with directly identifiable vocabulary: “trump”, “maga”, “gaetz”, “vote”...), a second profile related to the technical management of the platform itself (maintenance, moderation, network operations), and finally several accounts operating in niches entirely independent of these dynamics (video games, science popularization, animal-related news), whose rise occurs at much more random points in the year.

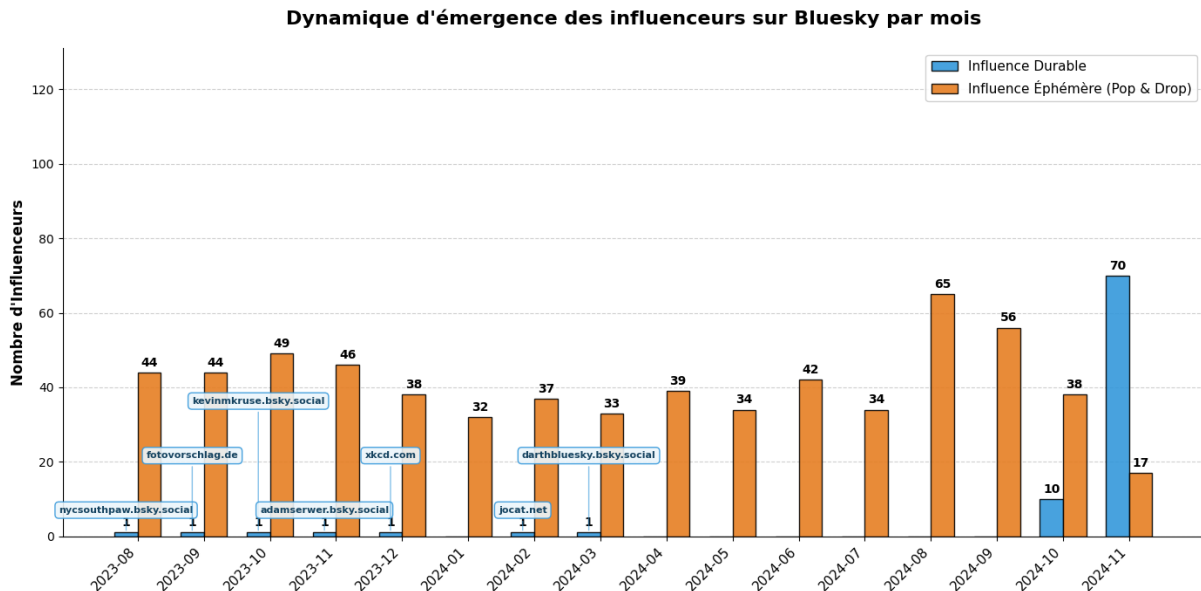


Figure 5.6: Monthly emergence dynamics of Rising Stars (blue: lasting influence; orange: ephemeral influence)

More broadly, matching the dominant terms of all ephemeral-rise accounts, month by month,

with real-world news makes it possible to trace an actual timeline of events that structured the network’s activity over the period studied (cf. Appendix, figure 6): near- continuous coverage of the Israeli-Palestinian conflict from October 2023 onward, several successive migration waves of non-English-speaking users (Brazilian in April 2024, then again in August-September 2024 – likely linked to the blocking of a competing platform in Brazil –, and Indonesian in June 2024), as well as the 2024 US electoral cycle culminating in November.

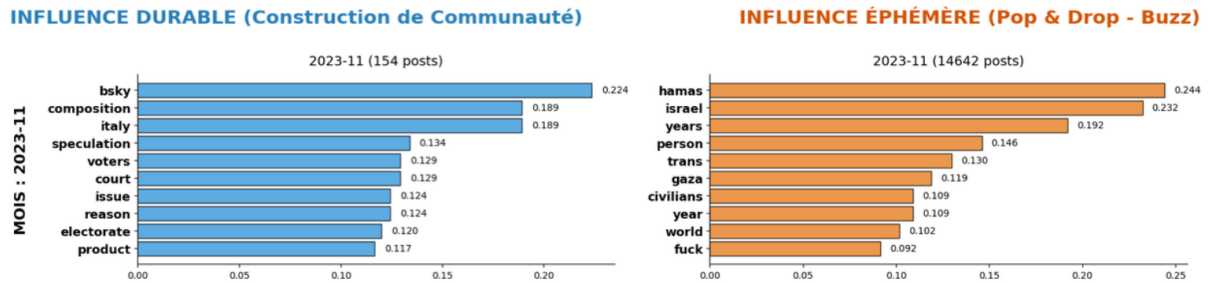


Figure 5.7: Example of the dominant themes of Rising Stars (TF-IDF analysis) for November 2023 (blue: lasting influence; orange: ephemeral influence)

Finally, an evaluation of the predictive power of structural algorithms (PageRank, K-Core, Betweenness, Cascade) with respect to the emergence of Rising Stars revealed their intrinsic limitations: these algorithms’ scores only rise at the very moment an account explodes in popularity, not beforehand, which appears to rule out any early prediction capability with the tools tested. PageRank and the Cascade score nevertheless allow, once the explosion has occurred, a clear distinction between lasting and ephemeral Rising Stars.

Évaluation Prédictive : Cohorte des influenceurs ayant explosé en 2024-10

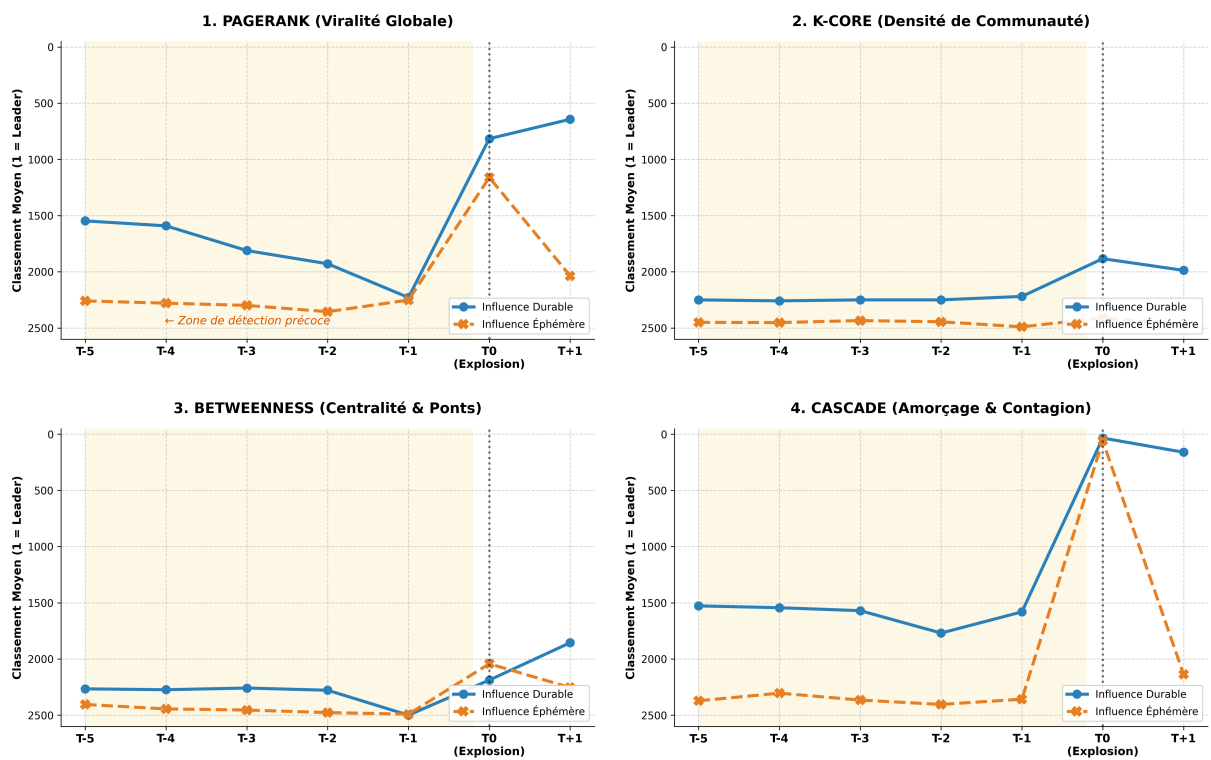


Figure 5.8: Evaluation of the predictive power of classical algorithms for Rising Stars (blue: lasting influence; orange: ephemeral influence)

Chapter 6

Problems Encountered and Solutions Implemented

Scaling up the processing pipeline to eighteen months of data was the main source of technical difficulty during the internship, due to the platform’s rapidly growing data volume (peaking around the November 2024 US elections):

- **Memory saturation when building the graphs:** the initial Pandas-based pipeline regularly saturated available RAM. *Solution:* migration to DuckDB, with explicit memory limits, disk-spill, and parallelism locks.
- **Betweenness centrality computation too costly** on large graphs. *Solution:* statistical approximation (Brandes’ algorithm [5]) with an adaptive sample count guaranteeing an error margin bounded at 5%.
- **Unmanageable cascade volume** (over 17 million discussion threads for a single month). *Solution:* an upfront filter excluding micro-discussions of three messages or fewer, or three distinct authors or fewer, eliminating 85 to 90% of the noise without losing relevant information.
- **“Ping-pong” bias** in the initial Spreading Power computation (repeated exchanges between two accounts artificially inflating measured depth). *Solution:* redefinition of depth and reach to count only distinct authors, and introduction of an anti-ping-pong filter keeping only the first relay of a pair of accounts.

Chapter 7

Project Management

The internship took place over thirteen weeks of active research, structured around regular individual meetings with Pr. Tsugawa. The project's progress was iterative rather than strictly sequential: several initial methodological approaches (for example, the first definition of the Spreading Power, or the initial choice to process everything with Pandas) proved inadequate and had to be revised, requiring flexible schedule management. A detailed Gantt chart of the project's actual progress is available in the appendix (Figure 1).

Chapter 8

Ecological Transition for Sustainable Development (TEDS)

8.1 Mobilan Simulation and Carbon Footprint of Mobility

The Mobilan tool was used when applying for the internship agreement to estimate the carbon footprint of the trip required to reach Japan, then filled in retrospectively with the actual data of my travel.

Mobilités

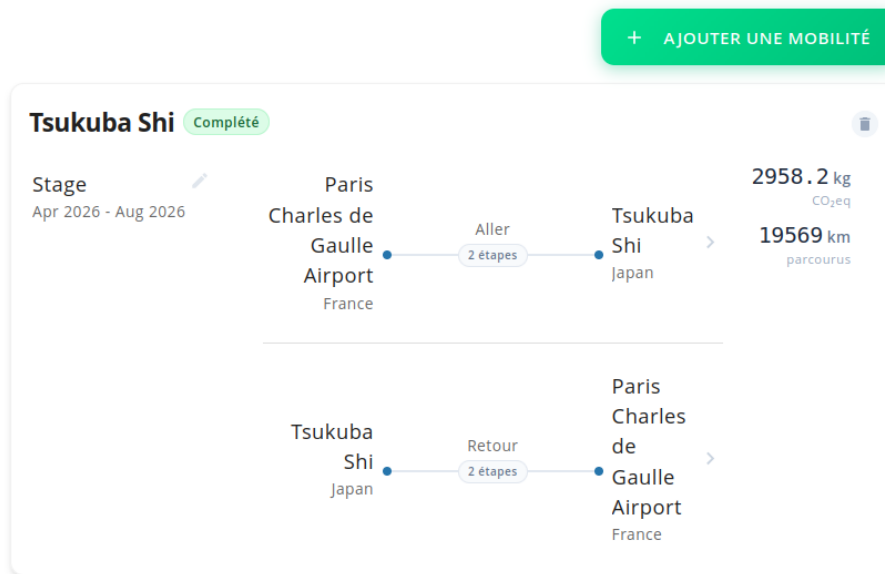


Figure 8.1: Screenshot of the Mobilan Mobility tab

The Mobilan tool did not allow me to travel differently for the main trip: alternatives to flying in order to reach Japan from France are in practice very limited (a boat journey would have represented a disproportionate travel time relative to the internship's duration). On the other hand, once on site, I systematically favored walking or cycling over the bus for my daily commutes, and the subway or train over flying for my longer-distance trips within Japan. Mobilan therefore mostly had a raising-awareness effect on my local mobility choices, more than on the international trip itself, which was difficult to avoid in the context of this internship.

8.2 CSR Policy of the Host Laboratory and Personal Reflection

Pr. Tsugawa's laboratory operates without fixed imposed working hours, each member freely managing their presence time (averaging 6 to 7 hours of work per day), with an individual check-in every two weeks rather than daily oversight. This organization, based on trust and autonomy rather than strict time control, struck me as a positive CSR policy on a human level: it respects each person's own working rhythm and avoids a culture of presence for presence's sake.

Access to the laboratory at night and on weekends, in the professor's absence, also raises a question of safety and of the institution's responsibility toward students who might come to work alone late at night. If I could do things differently, I would suggest setting up a simple presence-notification system (for example a simple notification when a student is working alone past a certain hour), which would preserve the current freedom of organization while reducing the risks associated with isolated work at unusual hours.

Chapter 9

Competencies

9.1 Specialty Competency: Designing Data-, Language-, Image- and Human-Centered Systems

This competency is justified by the following elements, developed in detail in the corresponding chapters of the report:

- **Collaborating with end users** (cf. Chapter 3, Problem Analysis and Action Plan): Pr. Tsugawa positioned himself as both the project’s sponsor and the end user of the models developed. My regular exchanges with him allowed me to understand his sociological hypotheses (criteria characterizing an influencer, relevance of the chosen ground truth) and to adapt the processing pipelines accordingly.
- **Choosing suitable algorithms and models** (cf. Chapter 2, State of the Art, and Chapter 5, Implementation): selecting and comparing graph-theoretic algorithms (PageRank, K-Core, Betweenness).
- **Properly structuring the data** (cf. Chapter 4, Technologies and Tools, and Chapter 6, Problems Encountered): setting up pipelines (DuckDB, identifier dictionaries, edge lists in Parquet) to transform billions of raw logs into exploitable graphs.
- **Validating the relevance of the approaches through evaluation** (cf. Chapter 5, section Result): creating an automated test bench evaluating each predictive model against the ground truth (overlap, Spearman correlation).
- **Complying with standards and recommendations (Ethics/GDPR)**: since the data is publicly accessible by design (Bluesky’s *firehose*), the ethical dimension here rests on understanding the GDPR exceptions related to academic research and on the responsible handling of this large-scale social data.

9.2 Cross-disciplinary Competency: Being an Engineer Who Collaborates and Shares Results

This cross-disciplinary competency is justified by the following six essential components, each linked to the corresponding chapters of the report:

- **By writing clear and accessible documents**: beyond this report itself, I kept a detailed weekly logbook throughout the internship documenting my hypotheses, methodological choices and their justifications (cf. Chapter 7, Project Management), which served as the basis for writing this report.

- **By communicating results with detailed argumentation:** every methodological choice was presented and argued during meetings with Pr. Tsugawa, for example the quantitative justification of the filtering threshold retained for the K-Core (cf. Chapter 5, section Static Structural Analysis) or the justification for the final choice of unweighted PageRank over more complex alternatives (cf. Chapter 5, section Result).
- **By adapting ad hoc tools for innovation, reflection, organization and planning:** faced with the progressive saturation of the existing infrastructure, I had to redesign and adapt the tools used on several occasions (migration from Pandas to DuckDB, replacement of NetworkX with a native BFS traversal, detachable `screen` sessions for long-running computations), revising the project's initial planning accordingly (cf. Chapter 6, Problems Encountered, and Chapter 7, Project Management).
- **By expressing my individuality within a group while respecting collective rules:** as an international intern within a Japanese laboratory, I proposed my own methodological choices (for example the successive iterations of the Spreading Power definition) while respecting the laboratory's collective framework (research directions defined together with Pr. Tsugawa).
- **By respecting intellectual property:** all reused concepts and algorithms (PageRank, Betweenness centrality, K-Core, as well as Pr. Tsugawa's prior work) are explicitly cited and referenced (cf. Chapter 2, State of the Art, and Bibliography).
- **By taking into account impacts on energy and greenhouse gas emissions, on biodiversity and on land resources:** this concern was reflected in two ways during the internship: on the one hand through the systematic optimization of the processing pipeline's computing resource consumption (memory, CPU time) (cf. Chapter 6, Problems Encountered), thereby limiting its energy consumption; on the other hand through my daily mobility choices in Japan, favoring walking, cycling and public transport over more emission-intensive modes (cf. Chapter 8, TEDS).

Chapter 10

Conclusion, Summary and Perspectives

10.1 Conclusion

This internship allowed me to conduct a complete, end-to-end research project in social network analysis: starting from raw, noisy, high-volume Bluesky activity logs, and ending with a validated, quantitative answer to the question of which classical graph-theoretic algorithm best identifies genuine influencers. The main finding (that a simple, unweighted PageRank restricted to the likes-only sub-network matches or outperforms more elaborate configurations (weighted multiplex graphs, filtered graphs, hybrid Borda scores) against the ground truth) shows that added algorithmic complexity does not automatically results into better predictive power, and highlights the importance of validating each candidate against an independently constructed ground truth rather than assuming a more sophisticated model performs best.

10.2 Summary

Starting from Bluesky’s raw activity logs, I built a complete pipeline to reconstruct interaction graphs and applied several classical graph-theoretic algorithms (PageRank, Betweenness centrality, K-Core) to identify influential accounts. In parallel, I built an independent, cascade-based ground truth (the Spreading Power) to evaluate these algorithms objectively, rather than relying on any single one by assumption. Systematically comparing every candidate against this ground truth over eighteen months of data showed that PageRank applied to the likes-only sub-network, without any weighting or filtering, offered the best trade-off between simplicity and predictive power – a simpler and more counter-intuitive result than initially expected. I also studied the temporal stability of the influencer core, distinguishing pillars, steady climbers and fading leaders, and analyzed the emergence of sudden-rising accounts, most of which turned out to cluster around the November 2024 US presidential election.

Beyond these results, I learned to iterate on a research question, to accept that an initial approach (for example an early definition of the Spreading Power, or the initial choice to process everything with Pandas) might prove incorrect and need revising, and more generally how graph theory can be mobilized to study a real social phenomenon. I also had to adapt to a new living and working environment in Japan.

10.3 Perspectives

A first avenue for further work concerns the early detection of sudden-rising accounts. The study conducted in this report showed that classical structural algorithms (PageRank, K-Core, Betweenness) only react at the very moment an account explodes in popularity, with no capacity for prior prediction. A finer approach would consist of combining machine learning and language

analysis: training a supervised classification model (for example using already-identified sudden-rising accounts as positive examples) on precursor indicators combining an account's still-modest structural position in the graph (rate of connection acquisition, recent evolution of its engagement) with the semantic content of its posts, matched against the topics currently identified as trending elsewhere on the platform. The goal would be to determine whether a still weakly influential account, whose discourse overlaps with a topic currently expanding elsewhere on the platform, presents a risk of imminent rise – an approach that would shift the question from mere structural characterization toward genuine early prediction capability.

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Appendix

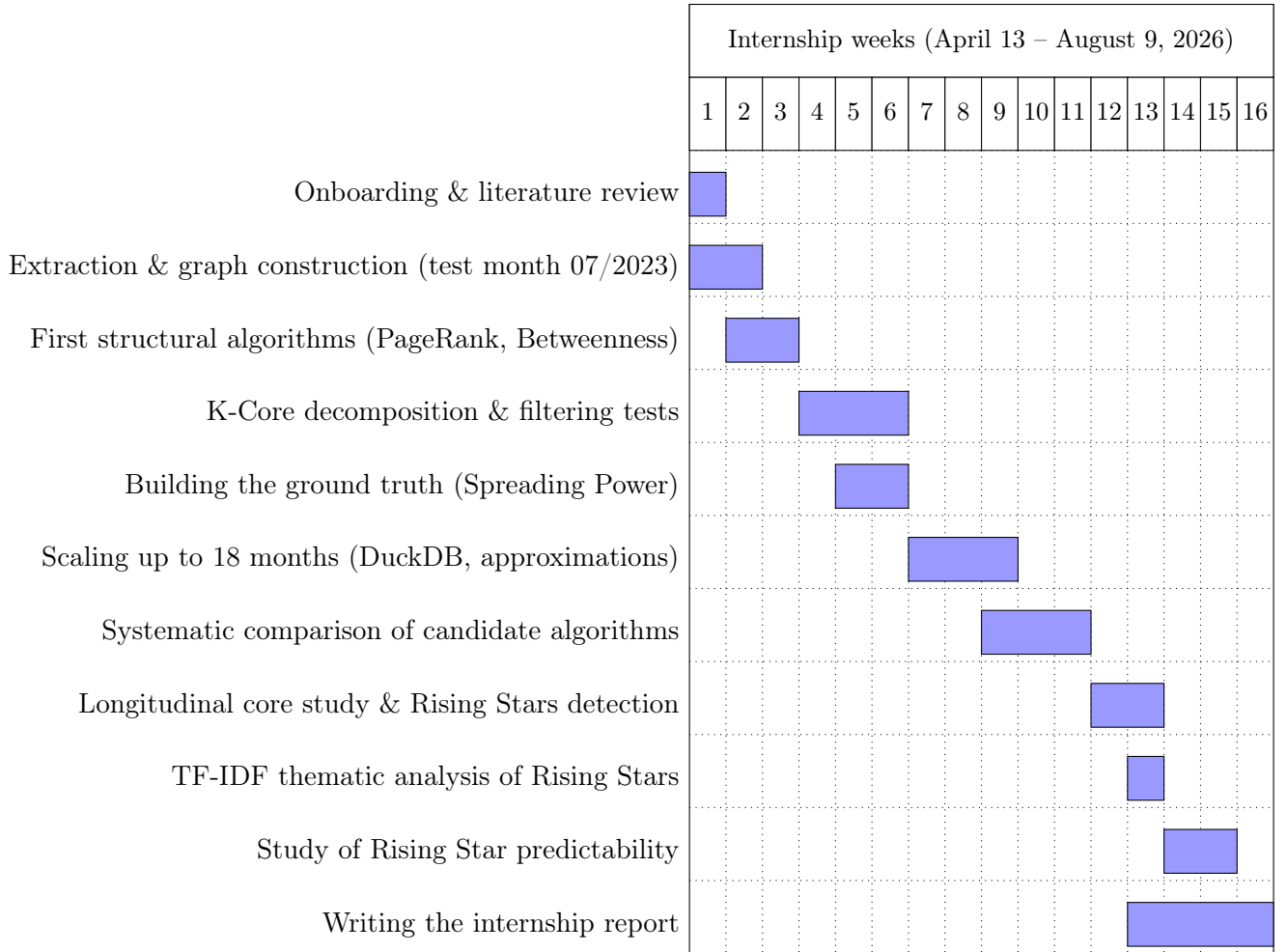


Figure 1: Gantt chart of the internship’s actual progress

OVERLAP TOP 2000 (%) - RÉSEAU : LIKE						
Month	PR (Unweighted)	PR (Weighted)	Betweenness	K-Core	Borda (Unw)	Borda (W)
2023-07	47.85	46.30	37.35	31.55	40.35	40.40
2023-08	41.90	39.65	30.00	23.10	33.60	33.40
2023-09	38.35	37.80	25.80	19.50	30.90	31.00
2023-10	39.05	37.00	22.35	16.35	28.65	28.45
2023-11	40.95	39.60	22.20	21.50	29.55	29.80
2023-12	41.65	40.10	22.75	18.00	29.70	29.50
2024-01	45.30	41.45	25.45	25.45	34.80	35.40
2024-02	36.40	33.50	20.95	26.00	34.70	34.65
2024-03	40.55	36.85	24.30	27.40	36.40	36.10
2024-04	44.35	39.25	24.85	26.90	36.60	36.00
2024-05	43.45	39.70	25.60	28.55	36.90	36.85
2024-06	43.90	39.90	24.05	26.05	33.70	34.35
2024-07	45.95	41.25	27.10	30.90	38.10	39.35
2024-08	46.10	42.85	20.55	24.25	29.60	30.00
2024-09	44.65	41.20	17.75	13.45	28.25	28.05
2024-10	38.15	37.00	16.60	17.60	31.00	30.50
2024-11	46.75	42.70	15.60	13.10	30.25	29.60
2024-12	50.55	44.85	15.90	20.80	35.20	35.85
=> MOYENNE GLOBALE DU RÉSEAU LIKE :						
PR (Unweighted)	43.10					
PR (Weighted)	40.05					
Betweenness	23.29					
K-Core	22.80					
Borda (Unw)	33.24					
Borda (W)	33.29					

Figure 2: Monthly comparison of algorithm prediction performance on the Only-Likes graph

OVERLAP TOP 2000 (%) - RÉSEAU : WEIGHTMAP						
Month	PR (Unweighted)	PR (Weighted)	Betweenness	K-Core	Borda (Unw)	Borda (W)
2023-07	46.55	42.40	37.10	29.80	41.60	41.85
2023-08	42.30	38.95	29.75	23.10	35.90	37.25
2023-09	37.95	34.55	25.30	21.50	32.85	34.55
2023-10	39.35	36.70	23.50	21.65	33.65	34.85
2023-11	41.05	36.95	22.85	23.90	34.45	36.25
2023-12	41.60	36.70	24.60	20.35	33.45	33.35
2024-01	45.10	39.70	24.80	26.80	37.90	40.00
2024-02	35.60	32.45	21.00	26.65	33.55	34.95
2024-03	39.25	34.20	23.55	28.10	37.55	39.80
2024-04	44.25	38.90	22.65	26.65	37.15	38.55
2024-05	43.05	37.65	24.75	27.35	37.55	40.05
2024-06	43.30	39.10	19.30	26.05	35.35	36.95
2024-07	45.70	40.40	26.65	30.55	38.55	40.85
2024-08	46.45	42.15	18.45	22.95	32.75	35.35
2024-09	46.65	41.35	15.70	12.60	32.90	33.05
2024-10	36.70	31.55	15.80	16.25	32.05	32.35
2024-11	46.20	39.75	21.20	19.80	39.70	40.65
2024-12	51.05	43.80	14.20	18.70	40.95	42.65
=> MOYENNE GLOBALE DU RÉSEAU WEIGHTMAP :						
PR (Unweighted)	42.89					
PR (Weighted)	38.18					
Betweenness	22.84					
K-Core	23.49					
Borda (Unw)	35.99					
Borda (W)	37.41					

Figure 3: Monthly comparison of algorithm prediction performance on the Multiplex graph (accounting for all metrics)

OVERLAP TOP 2000 (%) - RÉSEAU : REPOST						
Month	PR (Unweighted)	PR (Weighted)	Betweenness	K-Core	Borda (Unw)	Borda (W)
2023-07	43.25	40.65	34.95	30.90	40.50	40.40
2023-08	39.20	37.15	28.50	25.60	35.15	35.15
2023-09	34.80	33.15	24.35	22.50	32.45	32.40
2023-10	35.30	34.60	21.95	24.70	33.20	33.00
2023-11	37.00	33.95	23.50	25.60	34.25	34.15
2023-12	34.50	33.65	23.75	22.95	35.85	35.45
2024-01	41.25	37.50	25.85	29.00	39.10	38.75
2024-02	36.35	33.75	21.55	30.05	36.30	35.90
2024-03	40.00	36.55	24.75	30.65	40.20	39.55
2024-04	42.90	39.55	22.30	29.00	39.50	39.45
2024-05	41.45	38.55	24.15	28.40	39.55	39.25
2024-06	43.15	40.05	23.30	29.80	36.80	36.80
2024-07	44.20	40.60	26.20	30.35	41.60	41.60
2024-08	41.45	38.50	21.00	23.70	32.85	32.70
2024-09	37.15	35.95	15.90	12.10	30.35	30.65
2024-10	32.50	30.85	13.15	17.55	31.50	31.15
2024-11	38.90	36.20	14.80	21.15	36.20	36.10
2024-12	44.40	40.55	16.75	22.20	41.00	40.80
=> MOYENNE GLOBALE DU RÉSEAU REPOST :						
PR (Unweighted)	39.32					
PR (Weighted)	36.76					
Betweenness	22.59					
K-Core	25.34					
Borda (Unw)	36.46					
Borda (W)	36.29					

Figure 4: Monthly comparison of algorithm prediction performance on the Only-Repost graph

OVERLAP TOP 2000 (%) - RÉSEAU : FOLLOW						
Month	PR (Unweighted)	PR (Weighted)	Betweenness	K-Core	Borda (Unw)	Borda (W)
2023-07	31.30	31.30	21.55	8.90	31.75	31.65
2023-08	27.95	28.10	15.95	9.90	27.35	27.40
2023-09	26.90	27.10	13.60	9.30	29.55	29.65
2023-10	29.10	29.10	14.00	4.10	26.50	26.45
2023-11	28.10	28.05	13.15	4.35	26.75	26.80
2023-12	25.80	25.80	13.55	2.10	26.25	26.20
2024-01	28.60	28.50	13.50	2.25	25.90	25.95
2024-02	17.70	17.65	9.40	0.55	4.50	4.50
2024-03	22.65	22.85	11.60	0.20	16.10	16.00
2024-04	28.75	28.75	12.15	4.55	11.05	11.05
2024-05	29.15	29.05	11.35	0.65	15.75	15.70
2024-06	28.40	28.10	6.25	1.00	4.05	4.10
2024-07	30.95	31.10	11.75	0.85	24.80	24.75
2024-08	26.55	27.40	9.15	2.85	8.55	8.65
2024-09	37.25	37.25	11.25	0.65	14.45	14.55
2024-10	22.70	22.75	9.25	0.05	15.60	15.60
2024-11	41.50	41.45	19.10	5.50	37.45	37.40
2024-12	39.85	39.85	10.35	2.50	23.35	23.35
=> MOYENNE GLOBALE DU RÉSEAU FOLLOW :						
PR (Unweighted)	29.07					
PR (Weighted)	29.12					
Betweenness	12.61					
K-Core	3.35					
Borda (Unw)	20.54					
Borda (W)	20.54					

Figure 5: Monthly comparison of algorithm prediction performance on the Only-Follow graph

Timeline of events linked to influence spikes on Bluesky, Aug 2023 to Nov 2024

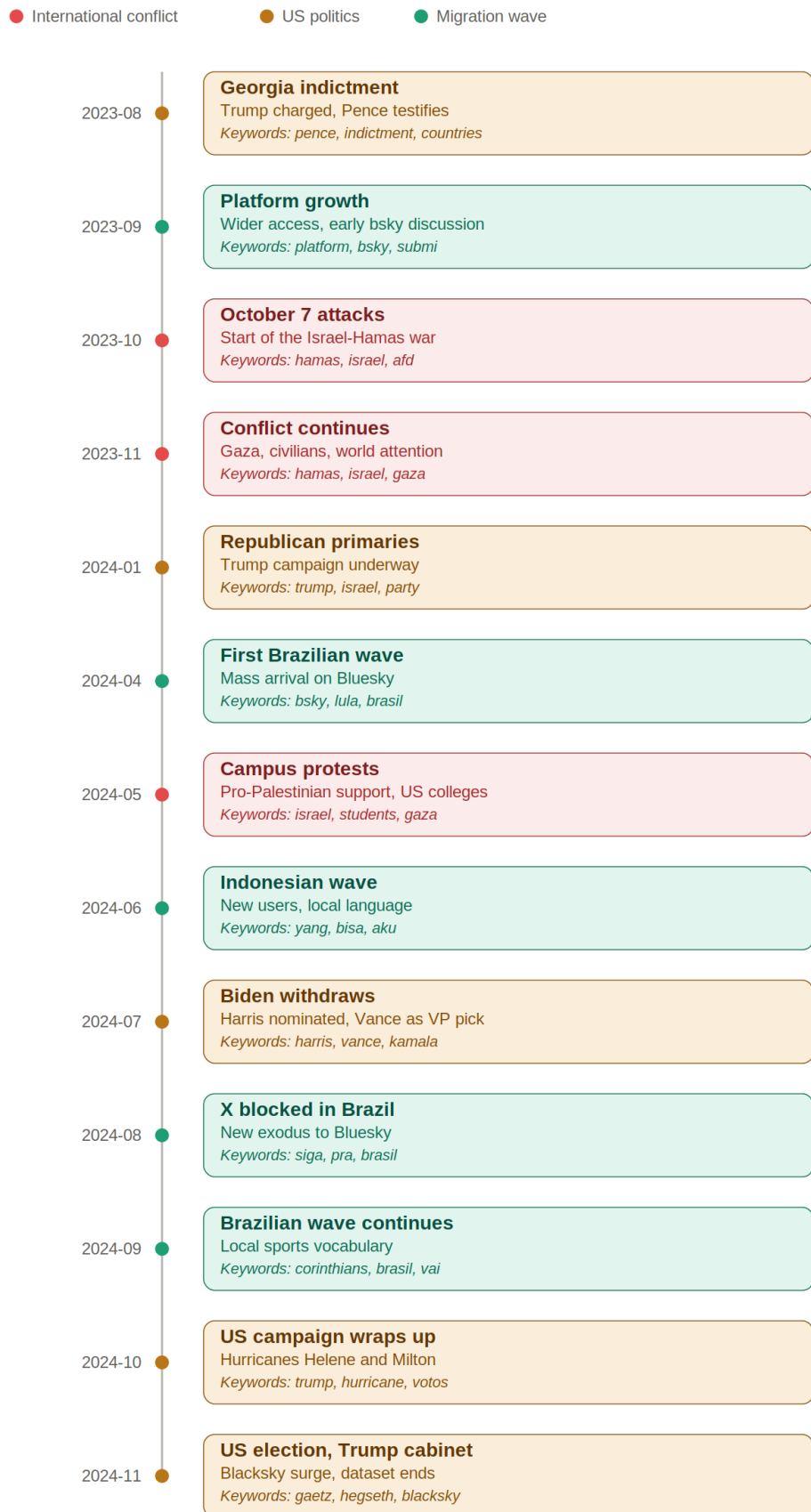


Figure 6: Timeline of real-world events identified through semantic analysis

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Characterizing and Identifying Influencers on Bluesky

Abstract:

For this internship, I joined the laboratory of Professor Sho Tsugawa at the University of Tsukuba, which specializes in social network analysis. My research topic focused on characterizing and identifying influencers on Bluesky. Starting from several months of raw JSON action logs, I built a complete pipeline to reconstruct interaction graphs and to apply various graph-theoretic algorithms to them, alongside a dynamic cascade-based ground truth. By systematically comparing these approaches over eighteen months of data, I established that PageRank applied to the like-only sub-network offers the best trade-off between simplicity and predictive power for identifying genuine cascade initiators, and I studied the temporal stability of the influencer core as well as the profile of accounts experiencing sudden rises in influence.